Kristiania

Study Guide: Seminar in Software Architecture (PIT9202)

Spring 2025

Introduction:

This course will introduce the candidates to the fundamentals of software architecture including design principles, common notations, and guidelines used to produce architectural design artefacts (styles, patterns, and frameworks). It will cover scenarios of software architectural design decisions and their impact on quality attributes of the software product (or service) including security, availability, scalability, and modifiability. It will also introduce the candidates to the methods, techniques, and tools used for describing software architecture and design rationale.

The candidates will learn the process, evaluation, refactoring, and knowledge management of architectural designs. In addition, the course will cover approaches and tools for designing and evaluating software architectures for the state-of-the-art technologies such as the Internet of Things (IoT), mobile computing, cloud-computing, and service-oriented architectures.

The objective of the course is to keep the candidate up to date with the state of the art and best practices in software architecture in building abstract as well as behavioral models to produce software that satisfy desired quality attributes. Additionally, it provides emerging trends and future challenges in software architecture.

Learning outcomes:

On completion of the course, the student has achieved the following learning outcomes, defined in terms of knowledge, skills, and general competence:

Knowledge

The candidate …

- is in the forefront of knowledge of theories and concepts behind software architectural design and masters the philosophy within the field of software architecture.
- can interpret theoretical quality of emerging paradigms within software architectures and contribute to development of the state- of-the-art knowledge, technologies and methods.
- can evaluate the expediency and application of software architectural styles, patterns, and design principles in research and scholarly development projects.
- can contribute to the development of new methods, tools and technics used for developing and refactoring software architectural artefacts.

Skills

The candidate can …

- independently formulate problems, plan and conduct research and scholarly development work in software architecture for applied information technology.
- carry out research and scholarly work of a high international standard in the systematic mapping of architectural patterns and software quality attributes in contexts including internet of things, mobile computing and cloud computing.
- handle complex academic issues and challenge established knowledge and practice by contributing with original research problem development in the field of software architecture for industry and public sector.

General competence

The candidate can …

- identify new relevant ethical issues and carry out his/her research with scholarly integrity in projects involving software architectures.
- communicate research and development in the area of software architecture through recognized international channels (e.g., IEEE and ACM).
- manage complex interdisciplinary assignments and projects involving software architecture and critically reflect upon the role and impact for software architecture.
- assess the need for, initiate and practice innovation in the field of software architecture with particular focus on green IT and sustainable IT solutions.

Degree: PhD in Applied Information Technology

Learning activities: Lectures, workshop, student presentation, and assignments

Teaching dates: March 3. - 7. 2025

Student workload

- Lectures and student guidance: 18 hours
- Self-study: 55 hours
- Preparation for presentation/discussion in class: 20 hours
- Assessment 40 hours
- Total: 133 hours

Compulsory assignments

- Non-verifiable coursework requirements: an oral presentation that must be approved
- Individual qualification: G/IG (approved/not approved)

- Execution: individual
- Verifiable (right of appeal): no
- Coursework requirements are to be handed in or conducted in accordance with information given by the lecturer and carried out within the duration of the course, as well as registered as approved/not approved at least two weeks before the exam/exam period.
- Approved coursework requirements grant students permission to take exams. Unapproved coursework requirements result in the student's withdrawal from the exam.

Examination

- Exam: Individual written home examination
- Duration: Semester
- Grading scale: The Norwegian grading system uses the graded scale pass or fail
- Weighting: 100 % of the overall grade
- Support materials: All support materials are allowed
- If graded fail, the candidate can resubmit the written home examination within 2 weeks from the grade is announced. The number of attempts for the exam is stated in the regulations.

Re-sit examination: N/A

Module structure:

Week	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
1	Self-study	Self-study	Self-study	Self-study	Self-study		
2	Teaching	Teaching	Teaching	Teaching	Teaching		
3	Self-study	Self-study	Self-study	Self-study	Self-study		

The topic is included in:

- PhD in Applied Information Technology

Course plan:

Date	Topics	Activity	Readings
Monday Mach 3, 2025	Software architecture fundamentals	Lecture Student activities: **Workshop**	Chapter 1-2 [1], articles
	Quality attributes of software systems		Chapter 3-14 [1], articles
Tuesday March 4, 2025	Architectural patterns and tactics	Lecture Student activities: **Workshop**	Chapter 4-14 [1], articles
	Documenting software architectures		Chapter 22 [1], articles

Wednesday March 5, 2025	The role of architecture in software projects	Lecture Student activities: **Workshop**	Chapter 24-25 [1], articles
	Evaluating software architectures		Chapter 21 [1], articles
Thursday March 6, 2025	Software architecture for the cloud	Lecture Student activities: **Seminar**	Chapter 17 [1], articles
Friday March 7, 2025	Emerging trends and future challenges in software architecture	Student activities: **Seminar**	Articles

Note on student activities:

- **Seminar** deals with student presentations and discussion.
- **Workshop** focuses on hands-on activities, skill-building, and practice.

References

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8. Patwardhan, A., & Patwardhan, R. (2016). XML Entity Architecture for Efficient Software Integration. *arXiv preprint arXiv:1606.07941*.
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